



BUILD WITH AI · CODE FOR COMMUNITIES

TRACK 3 · SMART HEALTH — AI-DRIVEN HEALTH CENTRE & SUPPLY CHAIN MANAGEMENT

Smart Health Centre

A real-time command centre for district health officials — and a 60-second daily report app for the PHC operator. Live on Google Cloud, in English, Hindi and Marathi.

Team **ai-rangers** · smart-health-20737125641.asia-south1.run.app · github.com/ai-rangers-team/Smart-Health-Center

THE PROBLEM

A district runs 30+ health centres on paper registers and phone calls

Primary Health Centres are rural India's front door to healthcare. But the people responsible for them are flying blind.

Stock-outs surface too late

A medicine shortage is discovered when a patient is turned away — not two weeks earlier, when a transfer or reorder could still have prevented it.

No live operational picture

Footfall, bed availability, staff attendance and diagnostics reach the district as monthly paperwork. Underperforming centres hide in the averages.

Tools aren't built for the users

PHC staff work in Marathi and Hindi on low-end phones. Dashboards built for English-speaking analysts never get adopted at the last mile.

One centre's surplus is another centre's stock-out. **The data to fix this exists — it just arrives a month late, on paper.**

Report once. The district sees everything, instantly.

- ✓ **PHC operator app** — the whole daily report (patients, staff, stock, beds, tests) takes under a minute on a phone, in the operator's own language.
- ✓ **District command centre** — live status of every centre: AI morning briefing, stock-out forecasts, alerts, performance flags, transfer recommendations.
- ✓ **Real-time, not batch** — every save recomputes forecasts, scores and alerts, and the dashboard updates in about a second. No refresh, no report cycle.

Built around one principle

Algorithms decide. Gemini explains.

Every alert and forecast is deterministic and auditable.

Generative AI is the communication layer — briefings, explanations, document reading — never the gatekeeper of a health decision.

Operator reports

60-second daily entry, or a photo of the restock invoice



System recomputes

Stock-out forecast, performance score, alerts — on every write



Firestore pushes live

District dashboard updates in ~1s via real-time listeners



Official acts early

Reorder, transfer stock, or intervene — days before the crisis

The district command centre

AI daily briefing, live alerts with days-to-stock-out, and transfer recommendations — one screen replaces a month of paperwork.

SHC

Smart Health Centre · District View

Health Dept.

English

Dr. A. Deshmukh

District Health Officer

Onboard a centre

Sign out

AI DAILY BRIEFING

Two centres face critical medicine stock-outs within 3 days. PHC Mulshi needs Paracetamol and PHC Haveli needs ORS urgently. A transfer from Pune Rural CHC is recommended for both.

AI flagged 3 stock-out(s) an average of 3 days early — time to act before shelves run empty.

Patients seen today

467

6 all centres

Critical alerts

2

Attention

Beds available

40

/ 110

All centres

6

District View

Active alerts

Critical

PHC Mulshi

Paracetamol 500mg will stock out in 3 days — reorder now

Resolve

AI Recommendations

Critical

Move 200 × Paracetamol 500mg from Pune Rural CHC to PHC Mulshi

Acknowledge

Generate redistribution plan

The same intelligence, in the official's own language

English, Hindi and Marathi across the entire product — and the AI writes its briefings, explanations and recommendations in the user's language, not just the labels.

SHC

Smart Health Centre · जिल्हा दृश्य
पुणे ग्रामीण जिल्हा · महाराष्ट्र आरोग्य विभाग

मराठी

Dr. A. Deshmukh
जिल्हा आरोग्य अधिकारी

नवीन केंद्र जोडा

साइन आउट

AI दैनिक आढावा

दोन केंद्रांमध्ये 3 दिवसांत गंभीर औषध तुटवड्याचा धोका आहे. PHC मुळशीला पॅरासिटामॉल आणि PHC हवेलीला ORS ची तातडीने गरज आहे. दोन्हीसाठी पुणे ग्रामीण CHC मधून हस्तांतरणाची शिफारस केली जाते.

AI ने 3 औषध-तुटवडे सरासरी 3 दिवस आधी हेरले — साठा संपण्यापूर्वी कृतीची वेळ.

आज तपासलेले रुग्ण
467
6 सर्व केंद्रे

गंभीर इशारे
2
लक्ष द्या

उपलब्ध खाटा
40
/ 110

सर्व केंद्रे
6
जिल्हा दृश्य

सक्रिय इशारे

गंभीर

PHC Mulshi
पॅरासिटामॉल 500 3 दिवसांत संपेल — त्वरित मागणी करा

निकाली काढा

AI शिफारसी

गंभीर

Pune Rural CHC कडून PHC Mulshi ला 200 × पॅरासिटामॉल 500 पाठवा

स्वीकारा

पुनर्वितरण योजना तयार करा

THE LAST MILE

A daily report that takes less time than a phone call

- ✓ **Designed for village PHC staff** — big tap targets, steppers instead of keyboards, works on low-end phones, installs as an app (PWA), tolerates patchy networks.
- ✓ **Scan the restock invoice** — when a delivery arrives, the operator photographs the invoice. Gemini reads it, matches line items to the centre's own medicine catalog, and pre-fills the quantities.
- ✓ **AI proposes, a human confirms** — extracted amounts land in a review screen, never straight into the database. One tap to accept, full control to correct.

Works in EN / HI / MR

Installable PWA

Gemini vision + structured output

मराठी

PHC Mulshi
Mulshi Taluka · मंगळ, ७ जुलै

साइन आउट

आज तपासलेले रुग्ण

- 0 +

भरलेल्या खाटा
8 पैकी 12

आज भरलेल्या

- 8 +

एकूण खाटा

- 12 +

आज उपस्थित कर्मचारी

आज उपस्थित डॉक्टर

- 0 +

एकूण डॉक्टर

- 2 +

तपासणी उपलब्धता

मलेरिया

उपलब्ध

उपलब्ध नाही

टीबी

उपलब्ध

उपलब्ध नाही

गर्भधारणा तपासणी

उपलब्ध

उपलब्ध नाही

मधुमेह

उपलब्ध

उपलब्ध नाही

एचआयव्ही

उपलब्ध

उपलब्ध नाही

अहवाल जतन करा आणि पाठवा

Medicine & vaccine stock
how much is left?

Scan restock invoice

Paracetamol 500mg

tablets

- 120 +

ORS Sachets

sachets

- 340 +

Iron + Folic Acid

tablets

- 450 +

Amoxicillin 250mg

tablets

- 300 +

Metformin 500mg

tablets

- 500 +

Algorithms decide. Gemini explains.

Health alerts must be reproducible and auditable. So the decision path is pure deterministic code — and Gemini makes it human.

Deterministic core — the decision path

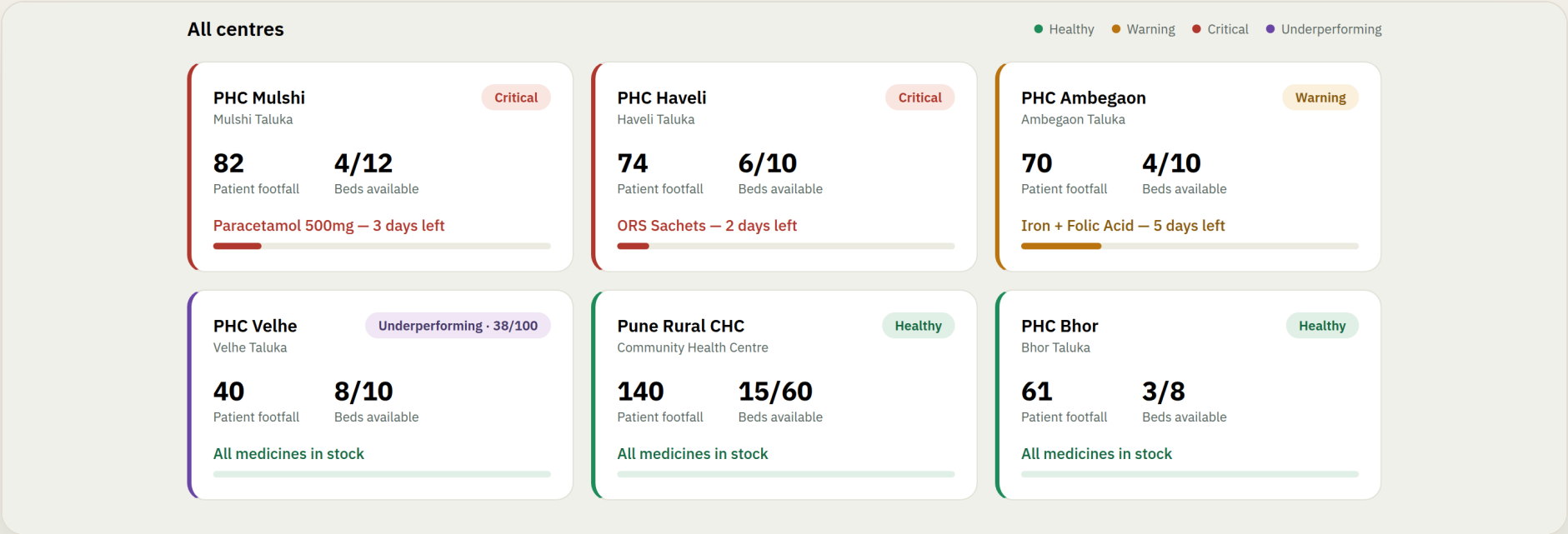
- ✓ Recency-weighted consumption forecasting (normalized EWMA) predicts the stock-out date of every medicine at every centre.
- ✓ Composite performance score across attendance, footfall, stock and diagnostics flags underperforming centres automatically.
- ✓ Redistribution engine matches surplus centres to deficit centres before stock-outs happen. Template-based alerts, zero hallucination risk.
- ✓ New centres are safe from day one: stock thresholds derived from one admin-friendly question — expected patients per day.

Gemini 2.5 Flash — the communication layer

- ✓ Morning district briefing written from live data, in the official's language, with cached last-good fallback so it never goes blank.
- ✓ Plain-language explanations of why a centre is underperforming and why a transfer is recommended.
- ✓ Invoice OCR: photo or PDF to structured stock quantities, constrained to the centre's own catalog.
- ✓ Never on the live write path — if Gemini is down or rate-limited, every alert, forecast and save still works.

From "we ran out" to "move 200 units by Friday"

Every centre card carries its most urgent forecast. The redistribution engine turns one centre's surplus into another's rescue — at zero procurement cost.



3 days

average early warning before a stock-out

7 / 14

day forecast bands for high and medium severity

~1 s

from operator save to district dashboard update

Not a prototype — a deployed, tested system



- ✓ **Security first:** role-based access via Firebase custom claims — a district admin and a PHC operator see only what they should.
- ✓ **60 automated tests** across forecasting, scoring, redistribution, alerting, auth and the invoice extractor.
- ✓ **CI/CD on every merge:** GitHub Actions deploys to Cloud Run with keyless Workload Identity Federation — no service-account keys anywhere.
- ✓ **Resilient by design:** cached last-good AI briefings, template fallbacks, deterministic writes even with Gemini fully unavailable.

Built for one district. Ready for a state.

Onboarding is one question

A new PHC gets safe stock thresholds from "roughly how many patients a day?" — and an entire district onboards from a single CSV: 30–40 centres, operators provisioned automatically.

The model rolls up

Centre to district today; district to state next. The same scoring, forecasting and redistribution logic works at every level of aggregation.

Integration roadmap

Designed to plug into e-Aushadhi (drug logistics), HMIS reporting and ABDM registries — replacing parallel data entry, not adding to it.

Every day of earlier warning is a queue of patients who did not get turned away. **The marginal cost of adding a centre is one row in a CSV.**



Rural healthcare that runs on data, not paperwork.

Live today. Tested. Trilingual. Built on Google Cloud.

Live demo

smart-health-20737125641.asia-south1.run.app

Source

github.com/ai-rangers-team/Smart-Health-Center

Team ai-rangers

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